

Chandrahas Nalabotu

✉ 101chandrahas@gmail.com | [101cn.github.io/portfolioCN-/](https://github.com/101cn)

EDUCATION

University of Leicester

2024 – Present

Mechanical Engineering MEng

- Year 1 and 2 - 1st

WORK EXPERIENCE / PROJECTS

Aerolabs CFD Analyst / Avionics Engineer – UAS Challenge

September 2025 – Present

CFD and Electronic Framework

- Ran 3D CFD simulations in STAR-CCM+ on a drone airframe, increasing lift coefficient from 1.0 to 1.6 by adding secondary flaps, enabling lower minimum flight speed.
- Reduced CFD simulation runtime from 6 hours to ~2 hours by tuning solver settings and convergence criteria under senior team member guidance.
- Configured flight controller optimising control delivery and sensitivity.

Innex1 Rover - Lunabotics 2026

September 2025 – Present

Software & Electrical Architecture

- Built a 5-topic ROS2 architecture with dedicated bridges for drivetrain, excavation and actuator subsystems, reducing controller input to rover response latency from 2.5s to 0.6s and enabling controller teleoperation via Pygame.
- Reduced live camera feed latency from 30s to ~1s at 40Hz through bandwidth reduction over a ROS2/WebSocket pipeline to mission control.
- Designed fail-closed motion safety logic with a 200ms stale-command timeout and stall-detection encoder verification to prevent uncontrolled spinning under low traction.
- Separated 60A motor power and 15A compute power domains with independent fusing and a fault-latching E-stop circuit, preserving telemetry and compute uptime during an E-stop event.

Surgical Instrument Carrier – Rover Delivery

September 2025 – March 2026

Hardware & Firmware Engineering

- Reduced electronics expenditure by ~30% by designing three distinct hardware subsystems (drive, indication, and servo) using off-the-shelf Arduino UNOs in EasyEDA rather than manufacturing custom PCBs.
- Achieved autonomous line-following capabilities by writing C++ control logic that continuously processed IR and Time-of-Flight sensor data to regulate dual brushed DC drive motors.
- Integrated a centralized firmware architecture across 3 separate hardware modules, successfully synchronizing the autonomous drive systems with the servo-actuated payload delivery mechanism.

Single Axis Stabilizer

September 2023 – May 2024

NEA Coursework

- Engineered a single-axis camera mount that actively counteracts external disturbance using IMU feedback while enabling continuous 360° rotation, controlled via Arduino.
- Reduced angular tracking error by 30% and eliminated dropped interrupts by refactoring open-source stabilization control code.
- Designed mechanical housing and PCB layout using Fusion 360 and EasyEDA, standardizing hole and via sizing to meet manufacturing tolerances.

TECHNICAL SKILLS

Software: Solidworks, Fusion 360, Excel, STAR CCM+

Coding / Frameworks: Python, Arduino, C++, MATLAB, Simulink, ROS2, Linux distributions

Manufacturing: 3D prints, lathe, laser cutter, soldering, standard workshop machining equipment